NILAM — CORE SYSTEM ARCHITECTURE

Land suitability · ArchAI2 house generation · Kerala construction cost estimation

MAIN PRODUCT FLOW

Inputs & Data

- Point · Radius · Brief
- Earth Engine · Sentinel-2
- SRTM · Soil · Hazards
- OSM · Places · Maps

Land Suitability

- 38-Feature Site Processor + Safety Gates
- TabPFN 3.5 — Tabular evidence
- TerraMind — Satellite imagery
- Cross-Attention + Calibrated Fusion
- Suitability Score · SHAP — Flat-land ranking · Amenities

ArchAI2 House Generation

- Selected Site + House Brief
- Qwen 3.5 9B Planning
- Layout + Geometry Engine
- Design Validation — 2D plan · 3D model

Cost Calculator

- Design Geometry
- District Rates
- Finish + Allowances
- Budget Breakdown

Final Outputs

- Land Assessment
- House Design
- Construction Estimate

DATA, TRAINING & STATE

Statewide Dataset & Model Training

1,400 Kerala points · 38 mapped features · dry and monsoon Sentinel-2 chips

TabPFN fitting · TerraMind fine-tuning · cross-attention training · fusion calibration

Project State

Selected plot · House brief

Design · Revisions

Kerala Cost Data

District base rates · Finish tiers

Services · External works · Contingency